

Two-Dimensional Hexagonal BX_3 ($X = P, As$) Monolayers as High-Capacity, Fast-Charging Anode Platforms for Lithium- and Sodium-Ion Batteries

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The rapid advancement of alkali-metal ion batteries demands robust anode platforms combining high specific capacities with rapid charge-discharge kinetics. Using first-principles density functional theory (DFT), we systematically evaluate two-dimensional (2D) hexagonal BX_3 ($X = P, As$) monolayers as high-performance dual-use anodes for lithium-ion (LIBs) and sodium-ion batteries (SIBs). Both metallic host architectures display strong thermodynamic affinities for Li^+ and Na^+ adsorption, favoring the hollow H_3 site through synergistic ionic charge transfer and orbital hybridization. Climbing image nudged elastic band (CI-NEB) calculations reveal low direct $H_3 \rightarrow H_3$ diffusion barriers: 0.40 eV (BP_3) and 0.26 eV (BA_{S3}) for Li^+ , and 0.26 eV (BP_3) and 0.19 eV (BA_{S3}) for Na^+ , confirming exceptional high-rate kinetics. Thermodynamic convex hulls establish maximum stable lithiation at Li_3BX_3 , yielding low average operating potentials of 0.39 V (BP_3) and 0.35 V (BA_{S3}) alongside theoretical specific capacities of 775 mAh/g and 341 mAh/g, respectively, with BP_3 doubling commercial graphite (372 mAh/g). For SIBs, multi-layer sodiation expands storage up to $Na_{15}BP_3$ and $Na_{12}BA_{S3}$, delivering ultrahigh capacities of 3875 mAh/g (BP_3) and 1365 mAh/g (BA_{S3}) at low voltages of 0.18 V and 0.15 V. Crucially, projected density of states (PDOS) analyses confirm that both frameworks preserve intrinsic metallic conductivity throughout all charging stages. These combined properties establish 2D BX_3 monolayers as outstanding, structurally resilient anode candidates for next-generation LIB and SIB energy storage technologies.

I. INTRODUCTION

Rechargeable batteries utilizing alkali metal ions have emerged as the paramount energy storage technology for modern electronics and grid-scale networks [1, 2]. Lithium-ion batteries (LIBs) form the bedrock of this infrastructure owing to their high working potential, low self-discharge rates, and superior energy density [3–5]. Despite these strengths, conventional LIBs face severe bottlenecks, including lithium resource scarcity, rising costs, transition metal extraction concerns [6], and thermal runaway risks exacerbated by flammable electrolytes [7–10]. Furthermore, the specific capacity ceiling of commercial graphitic anodes (372 mAh/g for Li^+) [11] restricts their ability to satisfy massive grid storage demands. Consequently, engineering alternative anode platforms remains a critical imperative [12].

To circumvent these challenges, sodium-ion batteries (SIBs) have emerged as a highly promising, resource-abundant alternative that readily adapts established LIB design principles [13, 14]. While the larger radius and heavier mass of the Na^+ ion pose kinetic challenges regarding solid-state diffusion, SIBs remain exceptionally viable for large-scale storage where cost-efficiency takes precedence over volumetric density [15]. Compared to emerging all-solid-state batteries [16, 17], SIBs benefit from simpler manufacturing routes and reduced material costs while maintaining highly

competitive electrochemical performance.

The realization of high-performance alkali-metal ion batteries is dictated by the choice of anode material, which governs capacity, rate capability, and cycling stability [18]. An ideal anode must facilitate rapid ion diffusion, preserve structural integrity, maintain electrical conductivity, and operate at low working voltages [19]. Traditional bulk anodes such as graphite, hard carbon, and metal oxides frequently suffer from sluggish kinetics, mechanical degradation, and severe volume expansion [20]. Consequently, research has pivoted toward two-dimensional (2D) nanomaterials. Atomically thin 2D sheets provide exceptional surface-to-volume ratios, flexible architectures, and abundant active adsorption sites that boost theoretical storage capacity while ensuring superior reversibility and rapid diffusion [21, 22].

Extensive density functional theory (DFT) screening has evaluated several distinct 2D families. Among elemental monolayers, pristine graphene delivers a double-sided lithium adsorption capacity up to 744 mAh/g [23], while phosphorene achieves 432.79 mAh/g for lithium storage with a remarkably low diffusion barrier of 0.09 eV [24]. Silicene offers lithium intercalation approaching bulk silicon (4200 mAh/g), albeit hindered by severe structural strain [25]. Transition metal dichalcogenides (TMDs) like MoS_2 push lithium capacities beyond 600 mAh/g [26], though structural transitions between their $1T'$ and $2H$ phases dramatically influence stability, and their poor intrinsic conductivity often requires carbonaceous hybridization [27]. Transition metal carbides and nitrides (MXenes), such as metallic V_2N , achieve exceptional lithium capacities (925 mAh/g) alongside near-zero diffusion barriers [28]. However, their surface functional terminations ($-F$, $-OH$) frequently obstruct transport, necessitating pris-

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tine or selectively functionalized surfaces for optimal performance [29, 30].

Amidst these candidates, boron-based nanomaterials have garnered intense interest. Elemental borophene demonstrates exceptional theoretical capacities for lithium storage (1239 to 1860 mAh/g) [31, 32], while carbon-hybridized BC_3 provides highly favorable adsorption sites for multiple alkali metal ions (Li, Na, K) [33]. Extending this paradigm to group III-V networks, boron pnictides demonstrate exceptional promise. Boron phosphide (BP) and its derivatives (BP_2 , B_3P) undergo favorable semiconductor-to-metal transitions upon ion adsorption, ensuring robust conductivity, delivering capacities up to 1691 mAh/g for both Li and Na storage, and providing ultralow diffusion barriers [34–36]. Two-dimensional boron arsenide (h -BAs) possesses excellent thermodynamic stability, delivering capacities of 522 mAh/g for both Li and Na ions, alongside low operating voltages [37]. Furthermore, BP and h -BAs monolayers function as highly efficient anchoring matrices in lithium-sulfur (Li-S) configurations, successfully mitigating the polysulfide shuttle effect while improving electronic conductivity [38, 39].

Building upon these established advantages, this work presents a comprehensive first-principles evaluation of BX_3 ($\text{X} = \text{P, As}$) monolayers as high-performance dual-ion anodes. Following recent computational investigations of the BP_3 framework for Na-ion storage [40] and the validation of intrinsic phase stabilities for both BP_3 and BAs_3 via density functional perturbation theory (DFPT) [41, 42], we significantly extend their functional scope. Utilizing DFT, we systematically map their single- and multi-layer alkali metal (Li and Na) adsorption configurations, multi-stage thermodynamic stabilities, and electrochemical voltage profiles. By rigorously evaluating the underlying charge transfer dynamics, localized chemical bonding, and surface migration trajectories, we detail the evolution of their electronic structures during sequential charging. Ultimately, we demonstrate that both BP_3 and BAs_3 monolayers preserve an intrinsically metallic architecture across their entire operational ranges, establishing them as highly promising, structurally resilient, and fast-charging 2D platforms for next-generation, high-capacity energy storage systems.

II. COMPUTATIONAL METHODS

First-principles calculations were performed within the framework of density functional theory (DFT) as implemented in the QUANTUM ESPRESSO package [43]. Electron exchange–correlation effects were treated using the generalized gradient approximation (GGA) parameterized by the Perdew–Burke–Ernzerhof (PBE) functional [44]. The interactions between valence electrons and ionic cores were modeled using optimized norm-conserving Vanderbilt (ONCV) pseudopotentials [45, 46]. Throughout the simulations, a plane-wave kinetic-energy cutoff of 80 Ry and a charge-density cutoff of 320 Ry were applied. For the integration of the Brillouin zone, a $12 \times 12 \times 1$ Monkhorst–Pack $\mathbf{k}$ -point mesh [47] was utilized in combination with a Methfessel–Paxton smearing

technique [48] featuring a width of 0.02 Ry.

To assess the electrochemical performance and phase stability of the proposed BX_3 ($\text{X} = \text{P, As}$) monolayers during alkali-metal intercalation, we evaluated their formation energy landscapes and open-circuit voltage (OCV) profiles. The thermodynamic stability of the intermediate intercalated phases was investigated by calculating the total formation energy (E_f) as a function of ion concentration (x):

$$E_f = E_{\text{tot}}(\text{M}_x\text{BX}_3) - E_{\text{tot}}(\text{BX}_3) - x \cdot E_{\text{tot}}(\text{M}_{\text{bulk}}), \quad (1)$$

where M denotes the intercalated alkali metal ($\text{M} = \text{Li, Na}$), and x is defined as the adatom concentration ratio relative to the pnictogen atoms ($x = N_{\text{adatom}}/N_{\text{X}}$, with $N_{\text{X}} = 8$ for the host supercell). Here, $E_{\text{tot}}(\text{M}_x\text{BX}_3)$, $E_{\text{tot}}(\text{BX}_3)$, and $E_{\text{tot}}(\text{M}_{\text{bulk}})$ represent the total energies of the intercalated structure, the pristine monolayer, and the corresponding bulk metallic phase (bcc Li or bcc Na), respectively. To determine the stable intermediate phases, a lower convex hull envelope was constructed across the E_f versus x space. Configurations residing directly on this hull are thermodynamically stable, whereas those lying above it are metastable or unstable, meaning they will spontaneously undergo phase separation into the two closest stable compositions.

The OCV profile as a function of lithium content was subsequently derived from these stable hull phases. For a lithiation step proceeding between two adjacent stable concentrations, x_{i-1} and x_i , the average operating voltage (V) relative to Li/Li^+ is calculated as:

$$V = -\frac{E_f(x_i) - E_f(x_{i-1})}{(x_i - x_{i-1})e}. \quad (2)$$

This thermodynamic approach yields a characteristic step-like voltage profile, where each plateau corresponds to a two-phase coexistence region during the transition between sequential stable lithiated phases.

III. RESULTS AND DISCUSSION

A. Electrochemical Storage Capacity, Voltage Profiles, and Structural Intercalation Pathways

To evaluate the capability and feasibility of the proposed BP_3 and BAs_3 monolayers as high-capacity dual-ion anodes for next-generation batteries, we systematically computed the stable intermediate phases during progressive Li and Na loading. The calculated thermodynamic convex hulls based on total formation energies, alongside the corresponding open-circuit voltage (OCV) profiles, are illustrated in Fig. 1, with key metrics compiled in Table I.

The thermodynamic stability of the intercalation processes was tracked by constructing formation energy hulls (Fig. 1a for lithiation, Fig. 1c for sodiation). The concentration x is defined as the ratio of adsorbed alkali atoms to pnictogen atoms ($x = N_{\text{adatom}}/N_{\text{X}}$), where $N_{\text{X}} = 8$ represents the total number of P or As atoms in the host supercell. During Li insertion, the total adsorption energy remains steadily negative down to

TABLE I. Calculated electrochemical parameters, including average adsorption energy (E_{ad}), total formation energy (Total E_{ad}), and open-circuit voltage (OCV), of Li- and Na-adsorbed BP₃ and BAs₃ monolayers at varying ion concentrations (x).

Ions	x	BP ₃			BAs ₃		
		E_{ad} (eV)	Total E_{ad} (eV)	OCV (V)	E_{ad} (eV)	Total E_{ad} (eV)	OCV (V)
Lithium (Li)	0.000	0.00	0.00	1.53	0.00	0.00	1.37
	0.125	-1.53	-0.19	1.53	-1.37	-0.17	1.37
	0.500	-1.05	-0.53	0.89	-0.95	-0.47	0.80
	1.000	-0.46	-0.46	0.26	-0.34	-0.34	0.23
	1.500	-0.38	-0.57	0.26	-0.32	-0.48	0.23
	3.000	-0.39	-1.17	0.26	-0.35	-1.06	0.23
	4.500	-0.17	-0.78	-0.26	-0.17	-0.77	-0.19
Sodium (Na)	0.000	0.00	0.00	1.62	0.00	0.00	1.49
	0.125	-1.62	-0.20	1.62	-1.49	-0.19	1.49
	0.500	-0.96	-0.48	0.74	-0.89	-0.45	0.70
	1.000	-0.54	-0.54	0.39	-0.54	-0.54	0.40
	1.500	-0.52	-0.78	0.39	-0.51	-0.77	0.40
	3.000	-0.48	-1.44	0.39	-0.48	-1.44	0.40
	4.500	-0.36	-1.62	0.11	-0.34	-1.53	0.07
	6.000	-0.29	-1.74	0.11	-0.27	-1.62	0.07
	7.500	-0.26	-1.95	0.10	-0.23	-1.73	0.04
	9.000	-0.23	-2.07	0.10	-0.20	-1.80	0.04
	10.500	-0.21	-2.21	0.10	-0.17	-1.79	0.03
	12.000	-0.20	-2.40	0.10	-0.15	-1.80	0.03
	13.500	-0.19	-2.57	0.10	-0.14	-1.89	0.02
	15.000	-0.18	-2.70	0.10	-0.13	-1.95	0.02

a concentration of $x = 3.0$. The appearance of clear cusps at $x = 0.5$ and $x = 3.0$ establishes these states as thermodynamically stable configurations against phase separation (Fig. 1a). However, increasing the Li content further to $x = 4.5$ causes an upward turn in the convex hull curve (yielding total adsorption values of -0.78 eV for BP₃ and -0.77 eV for BAs₃), indicating that over-lithiation beyond $x = 3.0$ is thermodynamically unfavorable. Conversely, the adsorption of Na ions remains an exothermic and highly favorable process across a much wider concentration range up to $x = 15.0$ (Fig. 1c). Configurations lying precisely on the solid lines of the convex hull represent the stable intermediate phases. As sodiation progresses, BP₃ demonstrates increasingly more negative formation energies compared to BAs₃, suggesting a stronger binding affinity for Na atoms and inherently greater thermodynamic stability in heavily sodiated states. These threshold behaviors are clearly confirmed by the step-wise OCV profiles presented in Fig. 1b (Li) and Fig. 1d (Na). Atomistically, these macroscopic voltage steps reflect the progressive and spatially distributed occupation of the host lattice.

During lithiation, the monolayers present initial voltage plateaus at 1.53 V for BP₃ and 1.37 V for BAs₃ ($x = 0.125$), driven by the exceptionally strong binding affinity of Li atoms at dilute concentrations. As the Li framework concentrates toward $x = 1.00$, the system must balance local binding energy against the increasing electrostatic repulsion generated by adjacent Li⁺ ions. To minimize this inter-ionic penalty, the adatoms adopt a spatially distributed arrangement across the host surface, which manifests electrochemically as a prolonged, flat OCV plateau of approximately 0.26 V for BP₃ and 0.24 V for BAs₃ spanning the capacity interval from $x = 1.00$

to $x = 3.00$.

The abrupt voltage drop observed beyond $x = 3.00$ is directly correlated with a critical transition in the storage mechanism. Once the primary surface capacity is fully saturated, subsequent Li insertion is forced into less energetically favorable configurations (Table I). The high energetic cost of over-lithiation is evidenced by a noticeable reduction in average adsorption energy (E_{ad}) from -0.39 eV to -0.17 eV for BP₃ and from -0.35 eV to -0.17 eV for BAs₃ as the concentration increases to $x = 4.50$. This penalty forces the OCV into the negative region (-0.26 V for BP₃ and -0.19 V for BAs₃). Because negative operating voltages cause dangerous lithium metal buildup, the maximum safe lithium capacity is limited to $x_{max} = 3.0$.

The sodiation process similarly begins with high initial voltages 1.62 V for BP₃ and 1.49 V for BAs₃ at $x = 0.125$ driven by strong host-adatom binding. As sodiation progresses up to $x = 3.00$, the system balances binding affinity against electrostatic Na-Na repulsion by maintaining a dispersed single-layer coverage, resulting in a dominant OCV plateau at approximately 0.4 V. Beyond $x = 3.00$, a transition in the storage mechanism occurs as monolayer coverage reaches capacity, necessitating the formation of secondary and tertiary metallic layers. Because these outer Na layers interact primarily with underlying sodium atoms rather than the BX₃ substrate, the average adsorption energy experiences a step-down reduction. This shift from monolayer adsorption to multilayer metallic clustering drives the final OCV plateaus down to 0.11 V for BP₃ and 0.03 V for BAs₃ at their respective maximum capacities of $x = 15.00$ and $x = 12.00$.

Crucially, the OCV remains strictly positive throughout the

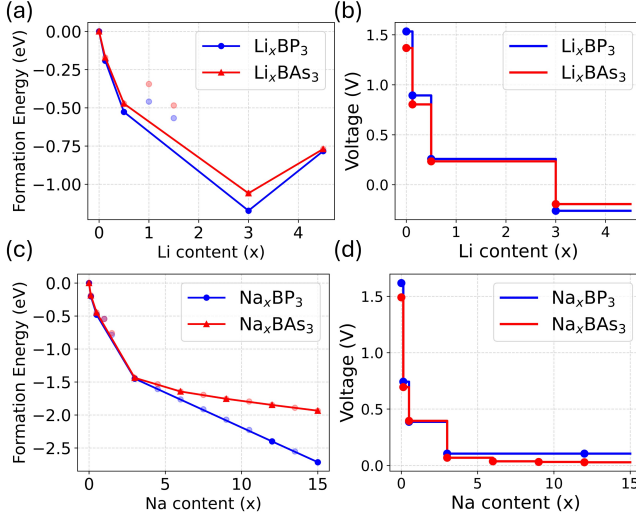


FIG. 1. Thermodynamic stability and electrochemical voltage characteristics of BP₃ (blue circles) and BAs₃ (red triangles) monolayers during sequential lithiation and sodiation. (a, c) Derived formation energy convex hulls as a function of alkali concentration x ($x = N_{\text{adatom}}/N_X$, where $N_X = 8$) for (a) lithium insertion (Li _{x} BX₃) up to $x = 4.50$ and (c) sodium insertion (Na _{x} BX₃) up to $x = 15.00$. Solid lines connect thermodynamically stable ground-state phases, while faintly shaded points indicate metastable or unstable configurations. (b, d) Corresponding stepwise open-circuit voltage (OCV) profiles for (b) lithiation relative to Li/Li⁺ and (d) sodiation relative to Na/Na⁺ across their full intercalation ranges.

full evaluated sodiation range. The fact that the adsorption energy remains strictly negative even at extreme sodiation states confirms that the multilayer sodiated phases are more energetically favorable than bulk sodium metal, guaranteeing a safe, dendrite-free operating mechanism near 0 V. Integrating the voltage plateaus yields exceptionally low average operating potentials: 0.39 V and 0.35 V vs. Li/Li⁺ for BP₃ and BAs₃, and 0.18 V and 0.15 V vs. Na/Na⁺. These values compare favorably with other 2D anode candidates such as elemental borophene (0.466 V) [31], SnSe₂ (0.66 V) [49], PC₆ (0.40 V) [50], tetragonal BN (0.35 V) [51], and graphene-like AlP₃ (0.28 V) [52]. Such low average potentials are highly advantageous as they maximize the overall full-cell output voltage when coupled with commercial high-potential cathodes, maximizing energy density while effectively mitigating thermal runaway risks.

To quantitatively assess the energy storage potential, the theoretical specific capacity (C) was calculated based on the fully lithiated and sodiated ground-state configurations using the relation:

$$C = \frac{x_{\text{max}} z F}{M}, \quad (3)$$

where x is defined as the concentration ratio of adsorbed alkali atoms relative to the pnictogen atoms in the host matrix ($x = N_{\text{adatom}}/N_X$), and x_{max} represents the maximum stable stoichiometry of adsorbed alkali atoms per formula unit (3.0 for Li; up to 15.0 and 12.0 for Na on BP₃ and BAs₃, re-

TABLE II. Comparison of theoretical specific capacities for BP₃ and BAs₃ monolayers against various 2D anode materials and conventional graphite for lithium- and sodium-ion batteries.

Type	Anode Material	Capacity (mAh/g)	Reference
Li-Ion	Silicene (multi-layer)	4200	[25]
	Borophene	1239–1860	[31, 32]
	V ₂ N MXene (metallic)	925	[28]
	BP₃ Monolayer	775	This work
	Pristine Graphene	744	[23]
	MoS ₂ (TMD)	≈ 600	[26]
	Phosphorene	432.79	[24]
	Conventional Graphite	372	[11]
	BAs₃ Monolayer	341	This work
	BP₃ Monolayer	3875	This work
Na-Ion	BP ₃ Monolayer	2325	[40]
	Dirac TiC	2015	[53]
	B ₃ P	1691	[35]
	BAs₃ Monolayer	1365	This work
	SiC ₂	1203	[54]
	MoPC	1157	[55]
	<i>h</i> -BAS	522	[37]
	Janus MoSSe	510	[56]
	Mo ₂ CO ₂ MXene	379	[57]
	Ti ₃ C ₂ T _{x} MXene	125	[58]

spectively). The term $z = 1$ is the valence charge number of the intercalant ion, $F = 26801$ mAh/mol is the Faraday constant, and M is the molar mass of the pristine BX₃ host framework ($M = M_B + 3M_X$). For the $2 \times 2 \times 1$ supercell model containing 8 boron atoms and 24 pnictogen atoms (B₈X₂₄), $x = 1/8$ (0.125) corresponds to a single adsorbed alkali atom. The individual atomic masses utilized are 10.81 g/mol for B, 30.97 g/mol for P, 74.92 g/mol for As, 6.94 g/mol for Li, and 22.99 g/mol for Na.

For lithium-ion storage, the terminal capacity limit of $x = 3.0$ corresponds to a high theoretical gravimetric capacity of 775 mAh/g for the BP₃ monolayer. As summarized in Table II, this value significantly outperforms conventional graphite (372 mAh/g) [11] and surpasses several prominent 2D anode candidates, including pristine graphene (744 mAh/g) [23], phosphorene (432.79 mAh/g) [24], and typical transition metal dichalcogenides such as MoS₂ (≈ 600 mAh/g) [26]. While lower than the ultra-high capacities of borophene (1239–1860 mAh/g) [31, 32], metallic V₂N MXenes (925 mAh/g) [28], and multi-layer silicene (4200 mAh/g) [25], the BP₃ framework circumvents the severe structural strain and restacking issues associated with extreme lithiation limits. Due to the heavier atomic mass of arsenic, the BAs₃ monolayer yields a reduced Li-storage capacity of 341 mAh/g.

For sodium-ion storage, the multi-layer capability allows the BP₃ monolayer to reach an ultrahigh theoretical specific capacity of 3875 mAh/g, while the heavier BAs₃ monolayer delivers a substantial 1365 mAh/g. This performance vastly exceeds typical MXenes (e.g., Ti₃C₂T _{x} at 125 mAh/g) [58], Mo₂CO₂ (379 mAh/g) [57], Janus MoSSe (510 mAh/g) [56], and the related *h*-BAS monolayer (522 mAh/g) [37]. Remarkably, it competes with or eclipses premier high-capacity

anomalies in the literature, including SiC₂ (1203 mAh/g) [54], the phosphorus-based B₃P monolayer (1691 mAh/g) [35], previous BP₃ SIB calculations (2325 mAh/g) [40], and Dirac TiC (2015 mAh/g) [53]. This exceptional dual-ion storage capability originates from the unique structural topology of the puckered BX₃ lattice, which can seamlessly accommodate dense, multi-layer alkali metal configurations symmetrically on both sides of the sheet.

To understand the stepped drops in the OCV profiles, we tracked the step-by-step filling sequence of alkali metals on the BX₃ monolayers (Fig. 2). Isolated adatom energy calculations show that the site stability hierarchy follows H₃ → B₁ → O₁ → H₄ for Na on both hosts, whereas for Li it follows H₃ → H₄ → B₁ → O₁ on BP₃ and H₃ → B₁ → H₄ → O₁ on BAs₃ (Fig. 4b, c). However, because the adjacent H₄ and B₁ sites lie in extreme spatial proximity, simultaneous occupation causes severe local Coulombic repulsion. To prevent this inter-ionic penalty, both systems dynamically bypass clustering at B₁ sites, establishing distinct effective filling pathways across the hosts: H₃ → H₄ → O₁ for Li and H₃ → O₁ → H₄ for Na.

This sequential filling directly drives the structural evolution and corresponding voltage plateaus. At dilute initial concentrations ($x = 0.125$), the first Li and Na ions fill the most stable hollow H₃ sites, producing strong binding and setting the high initial voltage plateau (Fig. 2a, g). As loading increases to $x = 1.50$, adatoms populate the secondary domains (H₄ for Li and O₁ for Na) to complete single-sided coverage on one face of the sheet (Fig. 2b–d for Li; h–j for Na). Further loading up to $x = 3.00$ symmetrically fills both faces of the monolayer with a dense, double-sided single layer (Fig. 2e, k).

Beyond double-sided single-layer saturation ($x > 3.00$), Li and Na adapt differently to their respective thermodynamic limits. For lithium, forced over-loading ($x = 4.50$, Fig. 2f) drives ions into high-energy tertiary positions, triggering an abrupt drop into unsafe negative working potentials. Conversely, continued sodium insertion (Fig. 2l–r) avoids this energetic penalty by transitioning into multi-layer metallic clusters symmetrically on both sides of the host, enabling substantially higher storage capacities while maintaining safe, positive operating voltages.

B. Alkali Metal Ion Diffusion Kinetics

The rate capability of next-generation ion batteries relies heavily on rapid metal-ion diffusion kinetics across the anode surface. To evaluate ion mobility, activation barriers and migration pathways were investigated using the climbing image nudged elastic band (CI-NEB) method in a $2 \times 2 \times 1$ supercell [59]. Minimum energy pathways (MEPs) were determined using 9 interpolated images for Path 1 and 11 images for Path 2. Because the H₃ hollow site represents the deepest thermodynamic energy minimum for isolated Li and Na adsorption, macroscopic transport is governed by inter-site H₃ → H₃ hopping. We evaluated two routes (Fig. 3a, c): a direct linear pathway (H₃ → H₃, Path 1) and an indirect pathway passing

through an intermediate hollow H₄ site (H₃ → H₄ → H₃, Path 2). The resulting energy profiles and activation barriers (ΔE) are presented in Fig. 3(b, d).

Path 1 is unambiguously the preferred diffusion channel in all evaluated systems due to the smoother energetic topology along the direct vector and the avoidance of the less favorable H₄ site. For Li diffusion, Path 1 yields smooth, single-barrier profiles of 0.40 eV (BP₃) and 0.26 eV (BAs₃), whereas Path 2 imposes double-peaked barriers of 0.61 eV and 0.64 eV, respectively. For Na diffusion, Path 1 exhibits exceptionally low barriers of 0.26 eV (BP₃) and 0.19 eV (BAs₃), compared to 0.44 eV and 0.47 eV for Path 2. Notably, the Na Path 2 profile exhibits a local minimum at the H₄ site, a feature absent in previous computational evaluations [40]. Comparing hosts, BAs₃ exhibits intrinsically faster kinetics than BP₃ for both Li (0.26 eV vs. 0.40 eV) and Na (0.19 eV vs. 0.26 eV) owing to its larger lattice constant and wider diffusion channels. These low Path 1 barriers (all < 0.50 eV) predict rapid room-temperature transport via the Arrhenius relation. For Li storage, the 0.26–0.40 eV barriers are highly competitive with graphite (≈ 0.40 eV) [11] and MoS₂ (≈ 0.25 eV) [26], while remaining higher than elemental borophene (< 0.01 eV) [31] or anisotropic phosphorene (0.09 eV) [24]. For Na storage, the 0.19–0.26 eV barriers indicate ultra-fast kinetics that compare favorably against reported 2D anodes including BP₂ (0.03 eV) [36], MoPC (0.06 eV) [55], Janus WSSe (0.07 eV) [60], SiP₂ (0.11 eV) [61], SnC (0.17 eV) [62], BSi (0.24 eV) [62], and BC₃N₃ (0.73 eV) [63]. These findings confirm that both architectures fulfill the prerequisites for high-power-density and rapid-charging applications.

C. Alkali Metal Adsorption Configurations

TABLE III. Convergence pathways from initial high-symmetry adsorption sites to final relaxed configurations and corresponding adsorption energies (E_{ad}) for Li and Na atoms on BP₃ and BAs₃ monolayers.

Ions	Final Site	BP ₃		BAs ₃	
		Initial Sites	E_{ad} (eV)	Initial Sites	E_{ad} (eV)
Li	H ₃	O ₂ , H ₁ , H ₃ , B ₂	-1.53	O ₂ , H ₁ , H ₃ , B ₂	-1.37
	H ₄	H ₄ , B ₃ , B ₄	-1.18	H ₄ , B ₃ , B ₄	-0.97
	B ₁	H ₂ , B ₁	-1.14	H ₂ , B ₁	-1.12
	O ₁	O ₁	-0.96	O ₁	-0.94
Na	H ₃	H ₁ , B ₂ , O ₂	-1.62	H ₃ , H ₁ , B ₂ , O ₂	-1.49
	B ₁	B ₁ , B ₃ , H ₂	-1.42	B ₁ , B ₃ , H ₂	-1.31
	O ₁	O ₁	-1.31	O ₁	-1.28
	H ₄	B ₄	-1.27	H ₄ , B ₄	-1.11

Prior to evaluating the electrochemical performance of the BX₃ (X = P, As) systems, the geometric structures of the pristine monolayers were fully optimized. The BAs₃ monolayer exhibits an optimized in-plane lattice constant of $a = 6.98$ Å and an intrinsic layer thickness of 1.39 Å. In comparison, the BP₃ monolayer features a slightly more compact structure, with a lattice constant of $a = 6.50$ Å and a thickness of 1.16

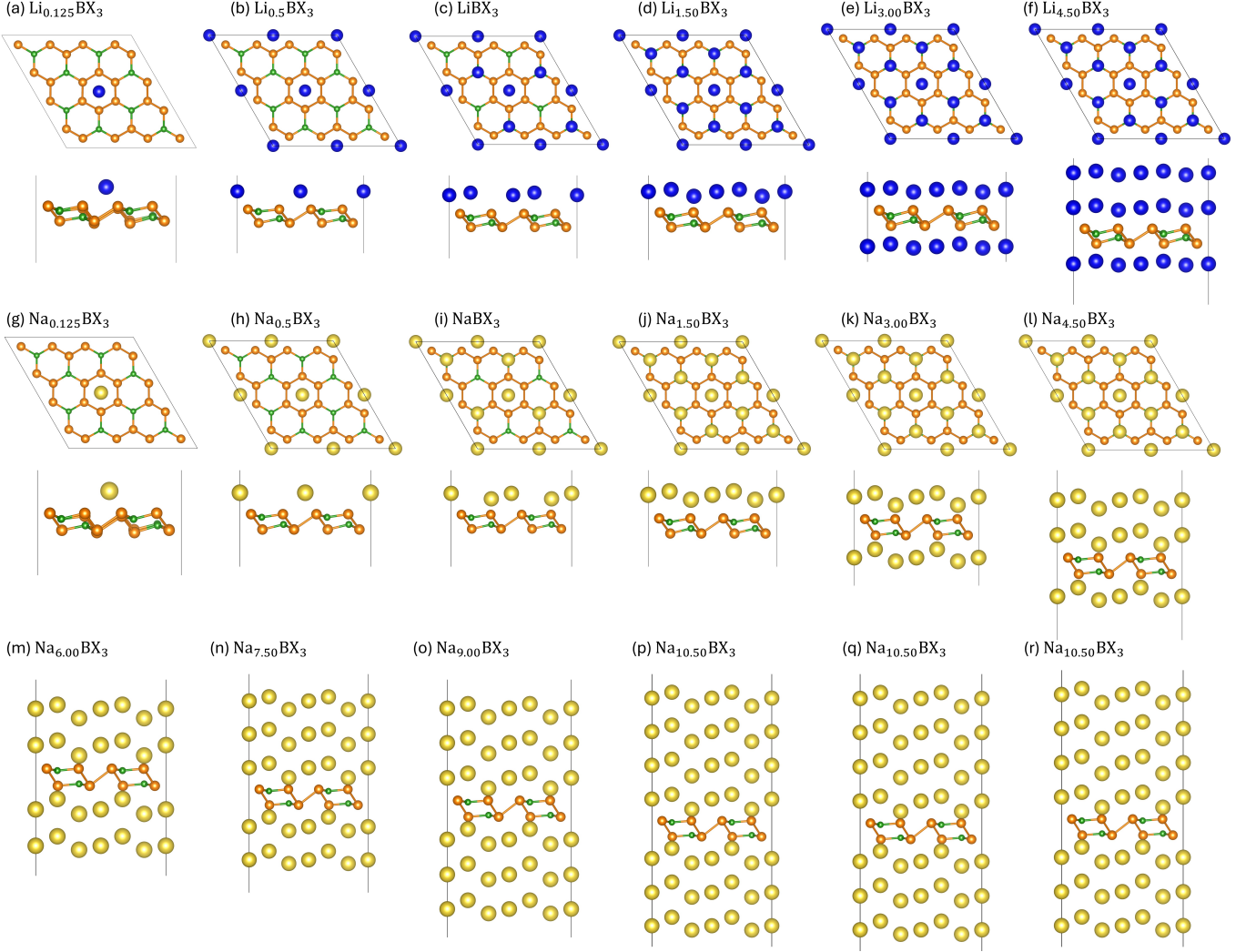


FIG. 2. Top and side views of the optimized geometric configurations during progressive alkali metal adsorption on the BX_3 ($X = \text{P}, \text{As}$) monolayers. The panels illustrate various concentration levels ($x = N_{\text{adatom}}/N_X$, where $N_X = 8$) explicitly denoted for both the lithiation (Li_xBX_3) and sodiation (Na_xBX_3) pathways. The structural evolution ranges from the initial dilute adsorption state at (a) and (g) $x = 0.125$ up to dense multi-layer metallic clustering at (f) $x = 4.50$ for Li and (r) $x = 10.50$ for Na. The green, orange, blue, and yellow spheres represent boron (B), pnictogen ($X = \text{P}, \text{As}$), lithium (Li), and sodium (Na) atoms, respectively.

Å. To eliminate any spurious interactions between periodic images along the out-of-plane direction, a large unit cell parameter of $c = 40.0$ Å was employed. This dimension ensures a substantial vacuum space of at least 20 Å, which is more than sufficient to accommodate the volumetric expansion of the systems even at their maximum theoretical multi-layer sodiation limits.

To identify the optimal alkali metal (Li and Na) storage mechanisms, we systematically investigated a comprehensive set of high-symmetry initial adsorption positions on both BX_3 surfaces. As illustrated in Fig. 4a, these are categorized into atop (O_i), bridge (B_i), and hollow (H_i) sites [40]. Following unconstrained structural relaxation, the various initial placements exhibited dynamic surface migration, ultimately converging into four unique and stable adsorption configurations for both materials: H_3 , H_4 , B_1 , and O_1 . As detailed in Ta-

ble III, the structural convergence pathways demonstrate a strong tendency for both Li and Na adatoms to migrate toward specific local energy minima. For example, initial adatom placements at the H_1 , B_2 , and O_2 sites consistently relaxed into the highly stable H_3 hollow site across all evaluated systems. Similarly, initial placements near the B_4 position consistently converged to the H_4 site, while the O_1 atop site remained stationary, retaining its initial configuration.

Evaluating the calculated adsorption energies (E_{ad}) of these converged configurations summarized in Fig. 4(b, c) and Table III reveals that the H_3 site is unambiguously the most thermodynamically stable anchoring point for both ions. It yields highly exothermic binding energies for Li (-1.53 eV on BP_3 ; -1.37 eV on BAS_3) and for Na (-1.62 eV on BP_3 ; -1.49 eV on BAS_3). Beyond the primary H_3 site, the relative stability ordering of the remaining sites varies slightly depending on

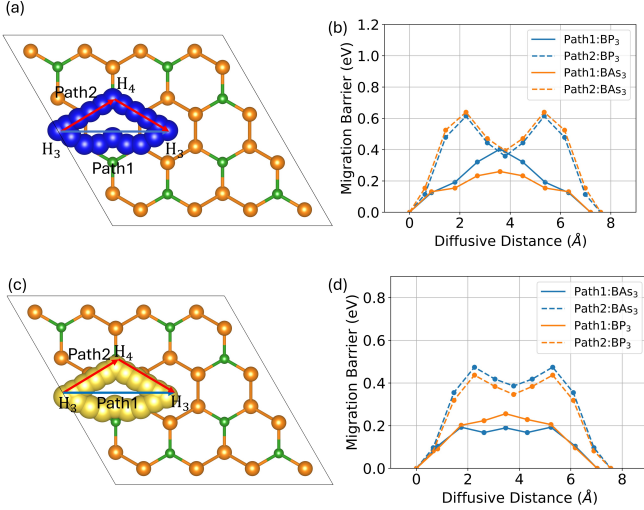


FIG. 3. Alkali metal ion diffusion kinetics on the BX_3 ($X = P, As$) monolayers. (a, c) Top views of the migration pathways between adjacent thermodynamically stable H_3 sites for (a) lithium and (c) sodium. Path 1 (blue arrow) represents the direct diffusion route, while Path 2 (red arrows) represents an indirect diffusion route passing through an intermediate H_4 site. The superimposed blue and yellow spheres illustrate the nudged elastic band (NEB) image trajectories for Li and Na, respectively. (b, d) Calculated minimum energy pathways (MEPs) and migration barriers for (b) Li and (d) Na migration. Solid lines correspond to Path 1 and dashed lines correspond to Path 2 for the BP_3 and BA_{S_3} systems.

the intercalant. For Na adsorption, both monolayers follow the exact same energetic hierarchy: $H_3 > B_1 > O_1 > H_4$. For Li adsorption, the ordering shifts to $H_3 > H_4 > B_1 > O_1$ on the BP_3 framework, and $H_3 > B_1 > H_4 > O_1$ on BA_{S_3} . Despite these minor mechanistic variations, all converged sites exhibit uniformly negative adsorption energies. This confirms a robust chemical affinity between the alkali metals and the puckered BX_3 networks, with the BP_3 monolayer consistently demonstrating slightly stronger ion binding compared to its arsenic counterpart.

D. Electronic Structure, Bonding Mechanisms, and Conductivity

To gain deeper insights into the underlying interaction mechanisms and chemical bonding nature between the adsorbed alkali atoms and the host systems (BP_3 and BA_{S_3}), charge density difference (CDD), Bader charge, and Crystal Orbital Hamilton Population (COHP) analyses were conducted. The spatial electron redistribution upon adatom insertion was evaluated using the relation:

$$\Delta\rho = \rho_{BX_3+metal} - \rho_{BX_3} - \rho_{metal}, \quad (4)$$

where the terms represent the charge densities of the adsorbed monolayer, the pristine host, and the isolated atom, respectively. The resulting spatial charge redistributions upon Li adsorption at the prominent H_3 and O_1 sites, and Na adsorption

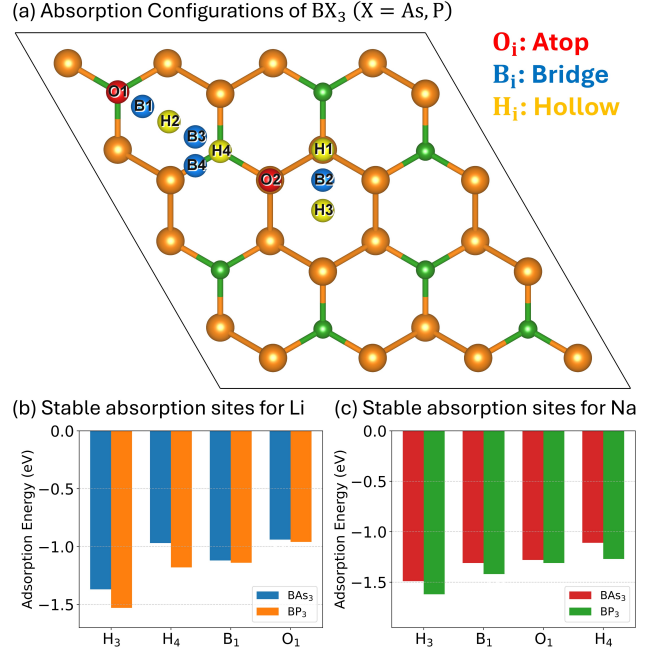


FIG. 4. Alkali metal adsorption characteristics on the BX_3 ($X = P, As$) monolayers. (a) Top view of the pristine BX_3 lattice illustrating the considered high-symmetry initial adsorption sites: atop (O_i , red), bridge (B_i , blue), and hollow (H_i , yellow). The green and orange spheres represent the boron (B) and pnictogen ($X = P, As$) atoms, respectively. Calculated adsorption energies (in eV) of the four most thermodynamically stable configurations after full structural relaxation for (b) lithium (Li) adsorption on BA_{S_3} (blue bars) and BP_3 (orange bars), and (c) sodium (Na) adsorption on BA_{S_3} (red bars) and BP_3 (green bars).

at the H_3 and H_4 sites, are illustrated in Fig. 5(a,b,d,e). A substantial charge transfer from the alkali atoms to the surrounding host network is clearly visible. The pronounced accumulation of electronic charge (cyan clouds) resides primarily on the adjacent pnictogen (P/As) atoms, accompanied by a noticeable charge depletion (yellow regions) tightly localized around the adatom.

Quantitatively, Bader charge analysis confirms that both lithium and sodium act as strong electron donors, revealing a prominent ionic character in the bonding mechanism. When localized at the primary H_3 site, the Li adatom loses $0.89e$ in the BP_3 system and $0.88e$ in the BA_{S_3} system. Similarly, the Na atom transfers approximately $0.84e$ to BP_3 and $0.83e$ to BA_{S_3} at the same site. Interestingly, this electron donation increases slightly when the ions occupy secondary sites; Li loses $0.92e$ (BP_3) and $0.91e$ (BA_{S_3}) at the O_1 site, while Na donates $0.88e$ (BP_3) and $0.87e$ (BA_{S_3}) at the H_4 site. From a purely electrostatic standpoint, this increased ionic character might imply that the secondary sites should exhibit the strongest binding energies. However, as established in the energetic analysis, the overall thermodynamic stability follows the opposite trend, indicating that ionic charge transfer alone is insufficient to explain the adsorption site preference, necessitating a deep evaluation of quantum mechanical orbital

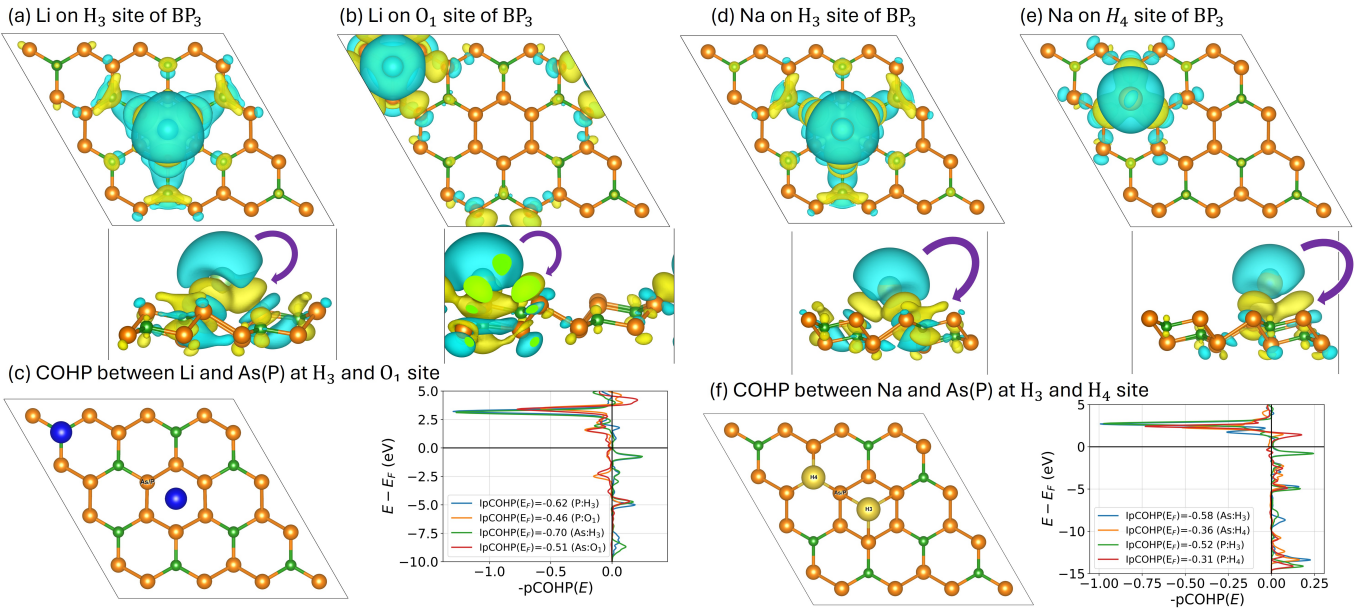


FIG. 5. Electronic structure analysis and chemical bonding characteristics of alkali metal adsorption on the BX_3 ($X = P, As$) monolayers. Top and side views of the charge density difference (CDD) isosurfaces for Li adsorption on BP_3 at the (a) H_3 and (b) O_1 sites, and Na adsorption on BP_3 at the (d) H_3 and (e) H_4 sites. The cyan and yellow regions represent electron accumulation and depletion, respectively. (c, f) Schematic representations of the relevant high-symmetry adsorption sites alongside the calculated projected Crystal Orbital Hamilton Population ($-pCOHP$) profiles for (c) Li-P and Li-As interactions at the H_3 and O_1 sites, and (f) Na-P and Na-As interactions at the H_3 and H_4 sites. The inset legends provide the integrated COHP (IpCOHP) values at the Fermi level (E_F), quantifying the relative covalent bond strengths.

overlap.

To resolve the specific covalent interactions between the adsorbates and the host atoms, we computed the projected Crystal Orbital Hamilton Population (pCOHP) implemented in the LOBSTER package. This method partitions the band-structure energy into bonding (positive $-pCOHP$) and antibonding (negative $-pCOHP$) contributions. As shown in Fig. 5(c,f), the interactions between the alkali metals and the neighboring pnictogen atoms are heavily dominated by bonding states below the Fermi level (E_F). The integrated COHP (IpCOHP) up to E_F serves as a quantitative metric for covalent bond strength where a more negative value denotes a stronger interaction. For lithiation, the Li-P interaction on BP_3 yields an IpCOHP of -0.62 eV at the hollow H_3 site, which is noticeably stronger than the value of -0.46 eV at the top O_1 site. The BAs_3 counterpart mirrors this trend, with the Li-As interaction exhibiting a robust bonding value of -0.70 eV at H_3 compared to -0.51 eV at O_1 . The sodiation data reveals an identical structural preference: the Na atom at the H_3 site exhibits profound covalent binding (-0.58 eV for Na-As and -0.52 eV for Na-P), while the H_4 site shows significantly weaker orbital hybridization (-0.36 eV for Na-As and -0.31 eV for Na-P).

Combining the ionic and covalent models provides a complete, synergistic picture of the adsorption mechanism. The highly electropositive alkali metal atoms bind to the surface through strong initial ionic interactions driven by near-complete valence electron donation, while their remaining electron clouds hybridize with the out-of-plane host orbitals to

form stabilizing covalent bonds. The superior thermodynamic stability of the H_3 site is structurally driven: its geometric position within the primary puckered ring allows adatoms to achieve highly efficient, multi-center orbital overlap with the surrounding P or As atoms. This optimal coordination maximizes covalent binding strength heavily outweighing any minor deficit in ionic charge transfer relative to secondary sites and cements H_3 as the primary anchoring point across both hosts and chemistries.

To evaluate electrical transport at maximum theoretical storage capacities, we computed the total and projected density of states (DOS) for the heavily intercalated configurations (Fig. 6). For both BP_3 and BAs_3 , the total DOS at maximum lithiation ($x = 3.0$, Fig. 6a,b) and heavy sodiation ($Na_{72}B_8X_{24}$, Fig. 6c,d) reveals a continuous, finite distribution of electronic states crossing the Fermi level (E_F). This confirms that the intrinsic metallic character of the pristine host frameworks is preserved and enhanced upon dense alkali-metal insertion. This metallic nature is driven by prominent peaks emerging at and above E_F , heavily dominated by the valence s orbitals of the intercalated ions (Li $2s$ and Na $3s$). Rather than remaining localized, the injected electrons populate the host conduction band and hybridize with host states, ensuring high intrinsic electrical conductivity during charge/discharge cycling. Maintaining continuous metallicity minimizes internal resistance and ohmic losses, facilitating the rapid electron transport required for fast-charging applications while reducing the need for conductive carbon additives. Additionally, uniform metallic charge delocalization helps screen local elec-

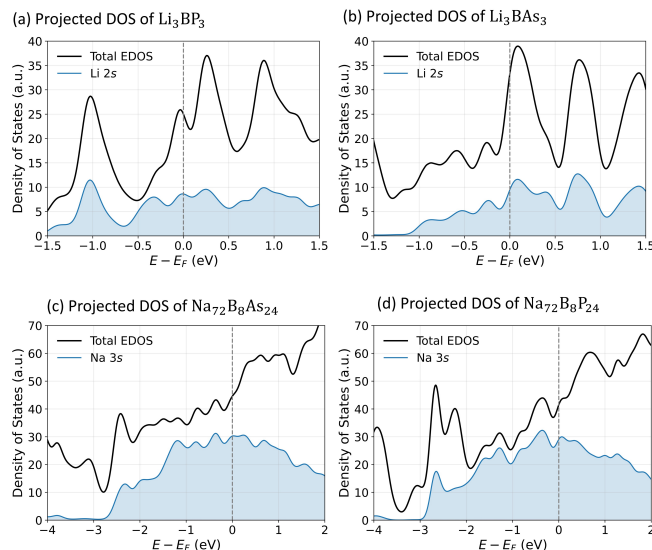


FIG. 6. Calculated electronic properties of the BX_3 ($X = P, As$) frameworks at high alkali metal concentrations. Total electronic density of states (EDOS, solid black lines) and projected density of states (PDOS) for the corresponding alkali metal s orbitals (Li $2s$ or Na $3s$, shaded blue profiles) for the lithiated (a) Li_3BP_3 and (b) Li_3BAS_3 phases, alongside the heavily sodiated supercell configurations of (c) $Na_{72}B_8As_{24}$ and (d) $Na_{72}B_8P_{24}$. The vertical dashed lines denote the Fermi level ($E_F = 0$ eV).

trostatic repulsions, preventing structural degradation or phase transitions during long-term cycling.

CONCLUSIONS

In summary, we evaluated 2D BP_3 and BAs_3 monolayers as anode materials for Li- and Na-ion batteries using first-principles density functional theory. Structural and energetic analyses show that both hosts share identical relaxation pat-

terns, with the hollow H_3 site serving as the most stable anchoring point for both alkali metals. This preference stems from a synergistic bonding mechanism combining substantial ionic charge transfer from the adatoms to the pnictogen framework with deep covalent orbital hybridization.

CI-NEB calculations reveal that both systems strongly favor direct linear hopping between adjacent H_3 sites, bypassing indirect routes to avoid electrostatic repulsion. This pathway yields ultralow diffusion barriers: 0.40 eV (BP_3) and 0.26 eV (BAs_3) for Li, and 0.26 eV (BP_3) and 0.19 eV (BAs_3) for Na. These low barriers ensure ultrafast charge/discharge kinetics suitable for high-power applications. Electrochemical assessments confirm high storage capacities within safe operating voltage windows. For lithiation, stability is bounded at Li_3BX_3 to avoid negative potentials, giving low average working voltages (0.39 V for BP_3 ; 0.35 V for BAs_3) and theoretical capacities of 775 mAh/g (BP_3) and 341 mAh/g (BAs_3). For sodiation, multi-layer metallic clustering enables stoichiometry up to $Na_{72}B_8X_{24}$, delivering ultrahigh capacities of 3875 mAh/g (BP_3) and 1365 mAh/g (BAs_3) with average open-circuit voltages of 0.18 V and 0.15 V, respectively.

Finally, density of states calculations confirm that intrinsic metallic conductivity is preserved across all intercalation levels for both ions. Electron injection from alkali s orbitals into the host conduction band ensures efficient electron transport at maximum capacity, reducing the need for conductive additives and screening local electrostatic repulsions. Overall, these findings position BP_3 and BAs_3 monolayers as promising, high-performance anode candidates for next-generation energy storage.

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